

3.1.5 Kinetics

The study of kinetics enables chemists to determine how a change in conditions affects the speed of a chemical reaction. Whilst the reactivity of chemicals is a significant factor in how fast chemical reactions proceed, there are variables that can be manipulated in order to speed them up or slow them down.

3.1.5.1 Collision theory

Content	Opportunities for skills development
Reactions can only occur when collisions take place between particles having sufficient energy. This energy is called the activation energy. Students should be able to: - define the term activation energy - explain why most collisions do not lead to a reaction.	

3.1.5.2 Maxwell–Boltzmann distribution

Content	Opportunities for skills development
Maxwell–Boltzmann distribution of molecular energies in gases. Students should be able to: draw and interpret distribution curves for different temperatures.	

3.1.5.3 Effect of temperature on reaction rate

Content	Opportunities for skills development
Meaning of the term rate of reaction. The qualitative effect of temperature changes on the rate of reaction. Students should be able to: use the Maxwell–Boltzmann distribution to explain why a small temperature increase can lead to a large increase in rate.	AT a, b, k and l PS 2.4 and 3.1 Students could investigate the effect of temperature on the rate of reaction of sodium thiosulfate and hydrochloric acid by an initial rate method. Research opportunity Students could investigate how knowledge and understanding of the factors that affect the rate of chemical reaction have changed methods of storage and cooking of food.
Required practical 3 Investigation of how the rate of a reaction changes with temperature.	

3.1.5.4 Effect of concentration and pressure

Content	Opportunities for skills development
The qualitative effect of changes in concentration on collision frequency. The qualitative effect of a change in the pressure of a gas on collision frequency. Students should be able to: explain how a change in concentration or a change in pressure influences the rate of a reaction.	AT a, e, k and i Students could investigate the effect of changing the concentration of acid on the rate of a reaction of calcium carbonate and hydrochloric acid by a continuous monitoring method.

3.1.5.5 Catalysts

Content	Opportunities for skills development
A catalyst is a substance that increases the rate of a chemical reaction without being changed in chemical composition or amount. Catalysts work by providing an alternative reaction route of lower activation energy. Students should be able to: use a Maxwell–Boltzmann distribution to help explain how a catalyst increases the rate of a reaction involving a gas.	